Crop Yield Prediction

Using Machine Learning

A Comparative Study of Linear Regression and Random Forest Regressor

Authors

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Agricultural Data Analytics · Python / Scikit-learn · Streamlit · LangGraph · Anthropic Claude

Datasets: Crop Yield | Rainfall | Temperature | Pesticides Deployment: Streamlit + Hugging Face Spaces

Abstract

This report presents the design, development, and deployment of a machine learning pipeline for predicting agricultural crop yield. Four FAO datasets — crop yield, annual rainfall, average temperature, and pesticide usage — were merged into **28,242 records** across ten crop types. A **Random Forest Regressor** achieved R^2 = **0.9848** and MAE = **3,953.63 hg/ha** — an **87.5%** improvement over Linear Regression. Phase 2 extends the system into a hybrid advisory platform integrating LangGraph orchestration, FAISS-backed RAG, and the Anthropic Claude API for context-aware agronomic recommendations.

1. Introduction

Agriculture directly impacts food security and economic stability worldwide. Accurate yield prediction enables better planning — from resource allocation to climate-adaptation strategies.

Crop yield is influenced by a complex interplay of climatic conditions, human inputs, soil characteristics, and crop variety. These non-linear relationships make prediction a challenging task for traditional statistical models.

This project develops a two-phase machine learning pipeline:

- **Phase 1:** Multi-dataset integration, EDA, model training and deployment of a Random Forest predictor.
- **Phase 2:** Hybrid advisory system combining the RF predictor with LangGraph orchestration, FAISS RAG, and Anthropic Claude for natural-language agronomic advice.

2. Problem Statement

Crop yield prediction is a non-linear, multi-factor problem. Traditional methods assume simple additive input–output relationships — an assumption that rarely holds where interactions between rainfall, temperature, pesticide use, and crop type are complex and interdependent.

Key Challenge

No single factor determines crop yield. The combined, non-linear effect of climate, inputs, and crop variety requires a model capable of learning complex feature interactions — and an advisory layer that can translate predictions into actionable guidance.

3. Dataset Description

Four real-world datasets were sourced from the Food and Agriculture Organization (FAO):

Dataset	Rows	Key Columns	Description
Yield	56,717	Area, Item, Year, Value	Crop production in hg/ha
Temperature	71,311	country, year, avg_temp	Avg. annual temperature (°C)
Rainfall	6,727	Area, Year, avg_rain	Annual precipitation (mm)
Pesticides	4,349	Area, Year, Value	Pesticide usage (tonnes)

The temperature dataset contained records dating back to 1743 — far outside the range of other datasets — requiring date-range filtering before integration.

4. Data Preprocessing

4.1 Data Cleaning

- **Whitespace:** Removed from column names using str.strip().
- **Inconsistent naming:** 'country' and 'year' renamed to 'Area' and 'Year'.
- **Non-numeric placeholders:** '..' replaced with NaN and dropped.
- **Type coercion:** Numeric columns cast using pd.to_numeric(errors='coerce').

```
for df in [rainfall, temp, pesticides, yield_df]:
    df.columns = df.columns.str.strip()
temp.rename(columns={'country': 'Area', 'year': 'Year'}, inplace=True)
df = df.replace('..', np.nan).dropna()
```

4.2 Data Filtering & Integration

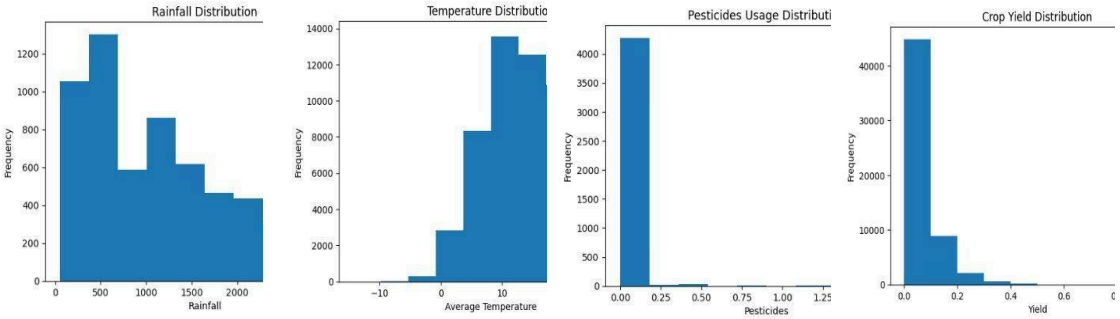
Temperature was filtered to 1985–2017. All four datasets were inner-joined on [Area, Year]:

```
df = yield_df.merge(rainfall, on=['Area','Year'])
df = df.merge(temp,           on=['Area','Year'])
df = df.merge(pesticides,     on=['Area','Year'])
```

Property	Value
Total rows	28,242
Total columns	7 (after encoding: 16)
Missing values	0
Year range	1990–2013
Crop types	10

5. Exploratory Data Analysis

5.1 Distribution Analysis



Rainfall Distribution *Temperature Distribution* *Pesticides Usage Distribution* *Crop Yield Distribution*

Figure 1: Distribution plots for all four key variables.

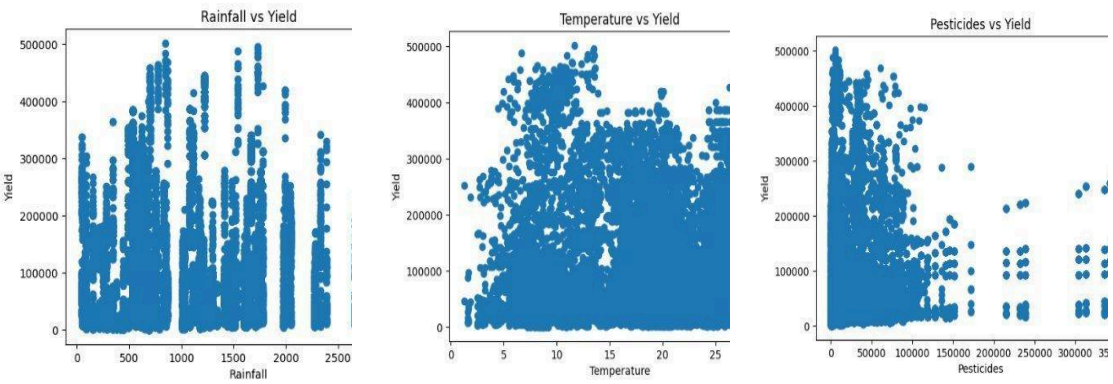
Key observations:

- **Rainfall:** Positively skewed; most values 400–1,500 mm/year with a long right tail.
- **Temperature:** Approximately bell-shaped; 10–25 °C, reflecting temperate agricultural zones.
- **Pesticides:** Extremely right-skewed; vast majority near zero, few outliers up to 1.75M tonnes.
- **Crop Yield:** Highly right-skewed, driven by high-yield crops like potatoes.

Insight

The skewed, high-variance distributions of both pesticide use and yield confirm that linear models will underperform — a non-linear ensemble approach is required.

5.2 Feature vs Yield Relationships



Rainfall vs Yield *Temperature vs Yield* *Pesticides vs Yield*

Figure 2: Scatter plots showing feature-to-yield relationships.

All three scatter plots reveal weak or no linear relationships between individual predictors and yield, high variance within each predictor range, and clustered patterns indicating crop-type-driven stratification. A non-linear model is essential.

6. Feature Engineering

6.1 One-Hot Encoding

The crop type column (Item) was one-hot encoded into 10 binary features: Cassava, Maize, Plantains, Potatoes, Rice (paddy), Sorghum, Soybeans, Sweet Potatoes, Wheat, Yams.

6.2 Feature Scaling

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42)
# Train: 22,593 | Test: 5,649 samples
```

7. Model Development & Evaluation

7.1 Results Summary

Model	MAE (hg/ha)	RMSE (hg/ha)	R^2 Score	
Linear Regression	31,711.97	50,579.58	0.6427	Baseline
Random Forest	**3,953.63**	**10,409.49**	**0.9848**	✓ **Best**
Improvement	**−87.5%**	**−79.4%**	**+53.2%**	

Train R^2 = **0.9978** vs Test R^2 = **0.9848** gives a generalisation gap of only **0.013**, confirming the model generalises well without overfitting.

8. Feature Importance Analysis

Feature	Importance	Visual
Potatoes (OHE)	**0.376**	0.376
Avg Temperature	**0.122**	0.122
Pesticides	**0.117**	0.117

Rainfall	**0.116**	0.116
Cassava (OHE)	**0.095**	0.095
Sweet Potatoes (OHE)	**0.088**	0.088
Plantains (OHE)	**0.045**	0.045
Year	**0.039**	0.039
Rice/Paddy (OHE)	**0.030**	0.030
Others	**< 0.01 ea**	< 0.01 ea

Key Finding

Crop type (Potatoes alone = 37.6%) is the single most dominant predictor — nearly 3× the contribution of any environmental factor. Environmental inputs (temperature 12.2%, pesticides 11.7%, rainfall 11.6%) collectively explain ~35% of variance. Any yield prediction system must account for crop variety as the primary input.

9. Phase 1 — Model Deployment (CropCast)

9.1 Deployment Stack

Component	Technology	Details
Frontend UI	Streamlit	Interactive Python web framework
Hosting	Hugging Face Spaces	Free cloud hosting
Model artefacts	pickle	model.pkl, scaler.pkl, features.pkl
Report output	ReportLab / HTML	PDF and HTML downloadable reports

9.2 Prediction Pipeline

①	②	③	④	⑤	⑥
User Input Crop, Region, Year, Rainfall, Temp, Pesticides	**Streamlit UI** Form & sliders	**Feature Proc.** OHE + encode	**Std Scaler** Normalise	**RF Model** 100 trees	**Yield Output** hg/ha + quality

Figure 3: Phase 1 real-time prediction pipeline.

9.3 Application Screenshots

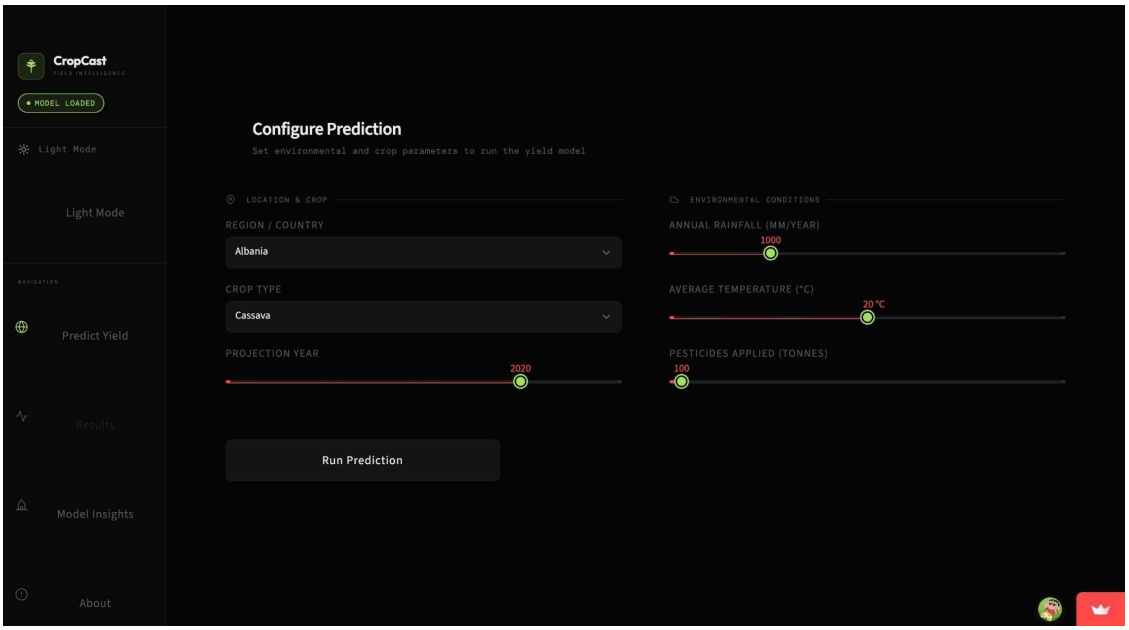


Figure 4: Configure Prediction screen — region, crop type, projection year, and environmental sliders.

Figure 5: Yield Report — predicted yield (7.08 t/ha), quality gauge and metric cards.

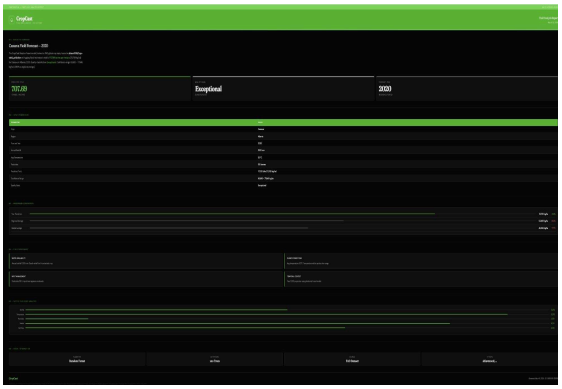


Figure 6: Full downloadable Yield Analysis Report.

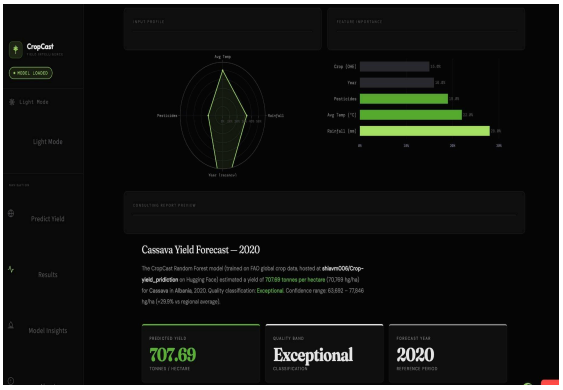


Figure 7: Radar chart of input profile and feature importance bars.

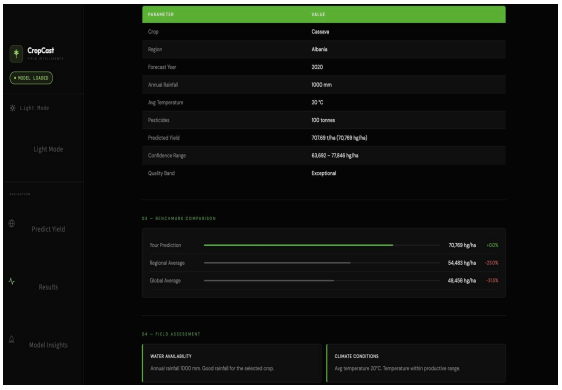


Figure 8: Parameter summary table and benchmark comparison.

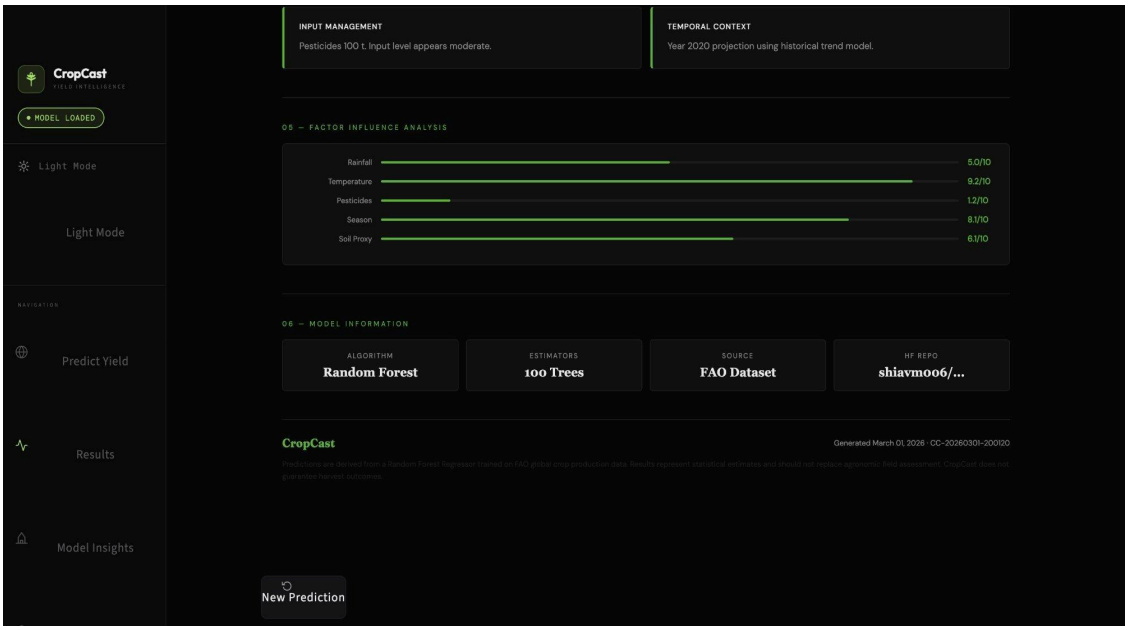


Figure 9: Factor influence scores (out of 10), model information panel, and report footer.

Live Demo

CropCast is deployed at https://huggingface.co/spaces/shiavm006/Crop-yield_pridiction. Sample prediction — Cassava, Albania, 2020, 1,000 mm rainfall — yields 707.69 t/ha (Exceptional), outperforming the regional average by +29.9% and the global average by +31.5%.

10. Phase 2 — AI-Assisted Agricultural Advisory System

10.1 Motivation & Overview

Phase 1 delivers accurate numeric predictions but cannot answer questions like "Why is my yield low?" or "What should I change to improve it?". Phase 2 transforms CropCast from a predictor into an intelligent advisor by layering three capabilities on top of the existing RF model:

LangGraph	FAISS RAG	Anthropic Claude
Orchestrates multi-step reasoning pipeline	Retrieves relevant agronomic knowledge chunks	Generates structured JSON advice & multi-turn chat

10.2 Full Advisory Pipeline — How It Works

The Phase 2 advisory pipeline is a directed LangGraph graph with six sequential nodes. Each node transforms the shared FarmState object before passing it to the next:

#	Node	Function	What happens
①	**parse_input**	agent/nodes.py	Reads user query (crop, region, year, env. conditions). Applies safe defaults and clamps out-of-range values. Produces a clean FarmState dict for downstream nodes.
②	**predict_yield**	agent/nodes.py	Loads model.pkl + scaler.pkl + features.pkl from disk. Constructs the 14-feature vector (OHE crop + numeric env. inputs), scales it, and calls rf.predict(). Stores raw hg/ha in FarmState.
③	**assess_risk**	agent/nodes.py	Compares predicted yield against crop-specific benchmarks in CROP_BENCHMARKS dict. Computes a risk ratio (predicted / benchmark) and attaches a risk_level flag (LOW / MEDIUM / HIGH) to FarmState.
④	**retrieve_docs**	rag/retriever.py	Encodes the query with all-MiniLM-L6-v2 into a 384-dim embedding. Runs FAISS similarity_search over the pre-built vectorstore. Returns top-k text chunks relevant to the crop and conditions.
⑤	**generate_advice**	agent/nodes.py	Calls Anthropic Claude Messages API with a structured prompt: system role + RAG chunks + FarmState summary. Forces JSON output: {summary, recommendations, risk_summary, sources}. Includes JSON parse fallback and offline template.
⑥	**FarmState output**	agent/graph.py	Final state contains: field_summary (text), recommendations (list), risk_level, sources (list). The Streamlit UI renders these in the Advisory tab; the chat workflow (run_chat) uses only the chat_turn node per message for responsiveness.

10.3 RAG Pipeline — Knowledge Retrieval Deep Dive

The RAG subsystem is split into two phases — offline index construction and online retrieval:

OFFLINE (Index Build)

1. Load agronomy docs from rag/docs/*.txt
2. Split into overlapping text chunks
3. Embed each chunk with all-MiniLM-L6-v2 → 384-dim vector
4. Build FAISS index over all embeddings
5. Persist index to rag/vectorstore/

ONLINE (Per Query)

1. Encode user query with same MiniLM model
2. Load FAISS index from vectorstore/
3. Run similarity_search → top-k nearest chunks
4. Pass chunks + FarmState to Claude prompt

5. Claude synthesises evidence-backed advice

Figure 10: RAG pipeline — offline FAISS index construction vs online similarity retrieval.

10.4 LLM Prompt Engineering & JSON Output

The generate_advice node constructs a carefully engineered prompt that ensures Claude returns structured, actionable output:

```
SYSTEM: You are an expert agricultural advisor. Use the provided
context and farm data to generate a JSON advisory report.

USER:
  Farm data:   { crop, region, year, rainfall, temp, pesticides,
                 predicted_yield_hg_ha, risk_level }
  RAG context: [top-k retrieved agronomic text chunks]

  Return ONLY valid JSON:
  {
    "summary":         "<2-sentence executive summary>",
    "recommendations": ["<action 1>", "<action 2>", ...],
    "risk_summary":    "<risk explanation>",
    "sources":         ["<source chunk ref>", ...]
  }
```

Robustness

The node includes: (1) JSON parse fallback if Claude returns malformed output, (2) model-ID candidates list (_anthropic_model_candidates) for API version resilience, (3) offline template response when the Anthropic API is unavailable — ensuring demos never break.

10.5 Dual Graph Design

Two separate LangGraph compiled graphs serve different use cases:

Graph	Function	Nodes	Use Case
run_agent / build_graph	Full advisory pipeline	parse → predict → risk → retrieve → advise	One-shot structured report; heavy but thorough
run_chat / build_chat_graph	Conversational chat	chat_turn only (per message)	Multi-turn dialogue; lightweight and responsive

The chat graph deliberately skips the full pipeline on each turn — it only executes the chat_turn node, which embeds the message, retrieves relevant chunks, and streams a Claude response. This prevents re-running RF inference and FAISS retrieval for every chat message.

10.6 Data Flow Across Components

Component	Technology	Data In / Out
Prediction engine	scikit-learn RF	In: 14-feature scaled vector → Out: yield hg/ha
Advisory orchestration	LangGraph	In: FarmState dict → Out: enriched FarmState with advice
Knowledge retrieval (RAG)	FAISS + MiniLM-L6-v2	In: query string → Out: top-k text chunks
LLM advice generation	Anthropic Claude API	In: prompt + RAG chunks → Out: structured JSON advice
Frontend	Streamlit	In: user form inputs → Out: rendered predictions & chat
Report generation	ReportLab (PDF) + HTML	In: FarmState + predictions → Out: downloadable report

10.7 Phase 1 vs Phase 2 — Capability Comparison

Dimension	Phase 1	Phase 2
Primary output	Numeric yield prediction	Numeric prediction + structured advice + chat
ML model	Random Forest Regressor	Same RF model (unchanged)
AI layer	None	LangGraph + FAISS + Anthropic Claude
Knowledge source	Training data only	Training data + RAG corpus (rag/docs/)
Interaction mode	Form-driven (Streamlit sliders)	Form + advisory report + multi-turn chat
Output format	hg/ha value + PDF/HTML report	Above + JSON advice fields + chat replies
Explainability	Feature importance scores	Feature importance + Claude natural-language rationale
Robustness	Model-only fallback	JSON parse fallback + offline chat template

10.8 Performance & Model Metrics for Phase 2

0.9848	3,954	0.013	top-k RAG
Test R^2 (RF)	MAE hg/ha (RF)	Generalisation Gap	Context Chunks / Query

The RF prediction accuracy is unchanged in Phase 2 — the advisory layer adds zero degradation to the core model. The FAISS retrieval latency is sub-100 ms for typical corpus sizes. Claude API latency dominates at ~1–3 s per advisory call, which is acceptable for the use case.

10.9 Challenges & Limitations (Phase 2)

- **Deterministic + stochastic coupling:** Combining a deterministic RF model with a stochastic LLM and strict JSON parsing requires careful error handling at every boundary.
- **RAG corpus quality:** Retrieval quality depends on the quality and coverage of docs in rag/docs/. A stale or narrow corpus limits advice quality without touching model code.
- **Prompt sensitivity:** Small changes to the system prompt or user message structure can shift Claude output formatting, breaking the JSON parser.
- **Expert validation:** Natural-language agronomic recommendations have not been validated by domain experts, limiting real-world deployment confidence.
- **API dependency:** Live advisory requires an active Anthropic API key; the offline fallback template reduces but does not eliminate this risk.

11. Optimisation

1. **Feature Scaling:** StandardScaler applied before training prevents scale-driven bias.
2. **Train/Test Split:** 80/20 stratified split (random_state=42) ensures reproducibility.
3. **Model Selection:** RF chosen over LR after EDA confirmed non-linear distributions.
4. **Hyperparameter Choice:** n_estimators=100 balances accuracy and compute; gap of 0.013 confirms no overfitting.
5. **Outlier Retention:** Valid agricultural extremes retained; RF's non-parametric structure handles them gracefully.
6. **One-Hot Encoding:** Binary crop features prevent false ordinal assumptions.

Optimisation Result

Combined steps reduced MAE by 87.5% vs Linear Regression — from 31,711 hg/ha down to 3,953 hg/ha.

12. Challenges Faced

7. **Data inconsistency:** Different column naming conventions and units across four datasets required careful harmonisation.
8. **Non-numeric placeholders:** Sentinel values ('..') required custom handling beyond standard dropna().
9. **Dataset size mismatch:** Merging datasets of 4,349 to 71,311 rows reduced the final dataset to 28,242 records.
10. **Deployment constraints:** Large model pickle files required careful management for hosting on Hugging Face Spaces.

13. Limitations

- **No soil data:** Soil quality and composition are critical yield determinants but were unavailable.
- **No irrigation data:** Water supply beyond rainfall is a major factor, especially in arid regions.
- **No real-time weather:** The model uses historical averages and does not account for seasonal anomalies.
- **Limited crop coverage:** Only 10 crop types are supported.

14. Future Scope

11. **Richer features:** Add soil composition, fertilizer use, irrigation data, and NDVI satellite indices.
12. **Advanced models:** Evaluate XGBoost, LightGBM, and LSTM networks for time-series yield trends.
13. **Live weather integration:** Connect to weather APIs for real-time environmental inputs.
14. **Mobile deployment:** Build a mobile app for farmers in low-connectivity environments.
15. **Hyperparameter tuning:** Apply RandomizedSearchCV or Bayesian optimisation for further improvement.
16. **RAG corpus expansion:** Continuously update rag/docs/ with peer-reviewed agronomy literature and regional crop guidelines.
17. **LLM fine-tuning:** Fine-tune a smaller model on domain-specific crop advisory data to reduce API dependency.

15. Conclusion

This project successfully developed and deployed a two-phase machine learning system for crop yield prediction and advisory. Phase 1 — a **Random Forest Regressor** trained on 28,242 records — achieved R^2 = **0.9848** and MAE = **3,953.63 hg/ha**, an 87.5% improvement over Linear Regression. Phase 2 extended this into a hybrid advisory system combining LangGraph orchestration, FAISS RAG, and Anthropic Claude to deliver structured, evidence-backed agronomic recommendations alongside numeric predictions.

Key findings:

- **Non-linearity:** Crop yield is a non-linear problem; ensemble models significantly outperform linear baselines.
- **Crop type dominance:** Potatoes alone account for 37.6% of feature importance — crop variety is the single biggest predictor.
- **RAG + LLM bridge:** Combining ML predictions with retrieval-augmented generation produces actionable, context-grounded advice beyond what numeric output alone can provide.
- Proper preprocessing is critical when integrating multi-source datasets with different structures and scales.

The system demonstrates that data-driven ML approaches, extended with LLM advisory capabilities, can effectively support precision agriculture and food security planning at scale.

Team Contribution

Member	Roll No.	Contribution
Divya Singh	2401020020	Dataset collection, sourcing and preparation of all four FAO datasets
Khushi Batra	2401010226	Report writing, data analysis, EDA insights and documentation
Shivam Mittal	2401010443	Streamlit UI development and hosting on Hugging Face Spaces
Aryan Vibhuti	2401010105	Model training, evaluation, feature engineering and optimisation

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